

MIDDLESEX UNIVERSITY LONDON

CST2550 Software Engineering Management and Development

Reset Coursework

Topic: Supermarket Management System (FreshMart Portal)

Module Code: 2550

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Submission Date: 04/07/26

1. Introduction

A web-based tool called the Fresh Mart Supermarket Management System was created to streamline and automate a supermarket's everyday operations. Conventional supermarket management frequently depends on manual record keeping, which can lead to issues including inadequate vendor management, delayed sales reporting, inaccurate inventory tracking, and invoicing errors. The objective of this system is to digitalize these procedures and increase overall efficiency.

The ASP.NET Core Razor Pages framework and C# programming language were used to create the application. Microsoft SQL Server and Entity Framework Core were utilized for database communication and data management.

Supermarket employees can use the system to process client transactions, manage inventory, add vendors, track sales history, swiftly search products, and keep an eye on dashboard statistics.

A login authentication mechanism was also implemented to block illegal access and ensure that only authenticated users can access administration functionalities.



FreshMart Portal

Supermarket Management Dashboard

Inventory

Manage products
and stock levels

Sales

Record supermarket
sales

Reports

View low stock and
summaries

Admin Login

Enter credentials to access features

Username

Password

Login

Demo: admin / admin123

2. System Design

The supermarket management system is built using a multi-layer web application architecture that connects the database, business logic, and front end.

The following elements are part of the system architecture:

- Product creation
- Product update
- Product deletion
- Sales processing
- Authentication validation
- Search operations
- Database communication

Layer of User Interfaces

- Dashboard
- Product Search
- Inventory Management
- Vendor Management
- New Sale Entry
- Sales History
- Login and Logout

ASP.NET Razor Pages was used to create the front-end user interface. Interaction between the user and the system is managed by this layer. It has pages like these:

Dashboard Inventory Management Product Search
Vendor Management Login and Logout New Sale Entry Sales History

Supermarket employees without technical expertise can simply utilize the application thanks to its straightforward and user-friendly interface.

Database Structure

The database layer uses SQL Server to permanently store all system data. Entity Framework Core manages communication between the database and the application.

This layer's primary function is to safely store product records, vendor information, and sales transaction history.

By separating display logic from data logic, this architecture facilitates system maintenance and future scalability.

3. Database Architecture

A relational database structure was used to develop the database design. To handle system data, three primary entities were established.

Stock Item Table

This table maintains all product information available in supermarket inventory.

Attributes:

- Product ID (Primary Key)
- Product Name
- Category
- Barcode
- Price
- Quantity Available

This table helps manage supermarket stock and allows products to be searched quickly using product name, barcode, or category.

CheckoutRecord Table

This table stores customer purchase transactions.

Attributes:

- Sale ID (Primary Key)

- Product Name
- Quantity Sold
- Price
- Total Amount
- Date of Sale

This table keeps a history of every transaction and helps generate sales reports.

4. Data Structures and Algorithms Used

The supermarket management system uses efficient data structures and algorithms to handle inventory, vendors, and sales records. The primary data structure used in the project is the List data structure in C#, which contains dynamic collections of items, vendors, and transaction data fetched from the database. This structure facilitates efficient retrieval and display of records on the web interface.

The major algorithm employed is a search algorithm that enables customers to search products directly using product name, barcode, or category without requiring other filter settings. The system validates user input, compares it against database entries, and shows matching results quickly. Additionally, the program makes extensive use of CRUD procedures (Create, Read, Update, Delete), which constitute the fundamental capability for handling sales transactions, vendor information, and inventories. These algorithms and structures increase the usefulness and efficiency of the system.

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Pseudocode of System Workflow

BEGIN

User Login

IF credentials are valid

Open Dashboard

Add/Search Product OR Record Sale

Update database

ELSE

Show invalid login

END IF

END

5. Testing Approach

The system was tested manually to ensure all major features worked correctly. Functional testing was performed on core features such as login authentication, inventory management, sales processing, and navigation. The following table shows the testing carried out during development.

Test Case	Action Performed	Expected Result	Actual Result	Status
Login Authentication	Enter valid username and password	User redirected to dashboard	Dashboard opened successfully	Passed
Invalid Login	Enter incorrect credentials	Access denied and remained on login page	Invalid login blocked	Passed
Add Product	Add new product with details	Product saved to database	Product stored successfully	Passed
Update Product	Edit product information	Updated details saved	Database updated correctly	Passed
Delete Product	Remove existing product	Product removed from inventory	Product deleted successfully	Passed

Product Search	Search by product name, barcode, or category	Matching product displayed	Correct results returned	Passed
Add Vendor	Enter vendor information	Vendor stored in database	Vendor added successfully	Passed
New Sale	Create a customer sale transaction	Sales recorded in system	Transaction stored correctly	Passed
Sales History	Open previous transaction records	Previous sales displayed	Records loaded correctly	Passed
Logout Function	Click logout button	User returned to login page	Redirect worked correctly	Passed

The testing procedure verified that the supermarket management system's main features operated as intended. Debugging was used to correct minor navigation issues discovered during development, producing a reliable and useful program.

6. Conclusion and Reflection

In summary, by offering a digital platform for inventory management, vendor administration, product searching, and sales tracking, the FreshMart Supermarket Management System was effectively created to enhance supermarket operations. The project's primary goal of using ASP.NET Core, C#, Entity Framework Core, and SQL Server to create a working web application while minimizing the need for manual data management was accomplished.

Throughout the development process, this project enhanced my knowledge of database design, CRUD operations, authentication systems, testing, and debugging while also giving me hands-on experience in full-stack web development. One of the biggest hurdles I encountered was fixing login and navigation-related problems, which required debugging and enhanced my problem-solving abilities. Even though the system still has drawbacks like rudimentary security and a dearth of sophisticated reporting capabilities, finishing this project boosted my confidence in software development and gave me useful experience creating practical applications.

References

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